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Drill Cuttings Characterization During In-Line Mud Logging Using Acoustic Sensors

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Abstract

This study presents a low-cost, non-invasive method for real-time diagnostics of multiphase flow in drilling systems using clamp-on MEMS microphones. The technique is based on the analysis of acoustic emission (AE) signals produced by the interaction of solid and water particles in a flow. The definition of characteristic grain size through solid type classification and the estimation of the solid mass flow rate are the two primary tasks that are addressed.

A hybrid signal processing pipeline is proposed for the classification of sand types. It combines Wavelet Decomposition (DWT) to extract statistical features of AE signals, which are then fed into a linear classification model. Welch's method is used to calculate the AE signal's Power Spectral Density (PSD) in order to estimate the mass flow rate. Band-wise root mean square (RMS) values of the PSD are used as regression features.

To validate the suggested method in a controlled laboratory setting, an experimental testbed was created. The classification model achieved a maximum F1-score of 0.917 after testing five different flow conditions, including a stream of pure water and four types of sand. For mass flow rate estimation, the method was tested across a range from 0 to 250 g/liter of solid, with predictions reaching a mean absolute error below 5 g/liter.

Introduction

Accurate assessment of solid mass flow rate and granulometry of multiphase flow is relevant in oil and gas industry operations for real-time detection of operational irregularities and downhole failure prevention. During drilling or production, the drill head or fluid injection causes a predictable rate of rock disintegration, and thus a theoretical prediction of the outflow of solid particles. Deviation from the expected and observed solid discharge can be a sign of hazardous subsurface conditions. Lower-than-expected solid flow rate can imply that cuttings are settling in the wellbore, resulting in sedimentation, increased torque, or even stuck pipe incidents. Conversely, an excessive solids production would be indicative of wellbore instability in the form of partial formation wall collapse or uncontrolled sanding – both of which are serious equipment integrity, production continuity, and safety threats. Monitoring solids transport in real time, therefore, is important for maintaining well conditions under control and ensuring safe operation.

A number of technologies have been developed to quantify solid flow rates and determine particle size distributions within industrial pipes, such as optical sensors, electrical resistance tomography, and nuclear-based techniques. Yet, non-invasive techniques like these are generally discovered to be inaccurate and too costly and maintenance-intensive for practical applications. Hence, the most accurate and common technique continues to be laboratory analysis of the samples collected at regular intervals. Though precise, the technique provokes serious operational challenges, including time delays between sampling and interpretation of the data, continuity loss in monitoring, and the incapacity to react to real-time behavioral changes. All these flaws render proactive decision-making unviable and laboratory tests unachievable for highly dynamic processes like drilling. Another encouraging option involves the use of acoustic emission (AE) signals to characterize the behavior of solid flow.

AE are the transient elastic waves generated by sudden release of stress in materials, typically by fracture or impact. In multiphase flow, AE may be generated by particle-particle and particle-wall impacts, turbulence-induced fluctuations, and flow instabilities. These frequency and amplitude modulation-rich acoustic signatures naturally contain information regarding particle size, mass, and dynamics – positioning them as an attractive indirect flow sensor. In the investigated flow regime in this research – water and sand slurry in a curved pipe – AE is primarily caused by solid particles impacting and sliding along pipe walls, especially in regions of curvature where particle trajectories are diverted by inertial forces. The particle and pipe wall interactions produce cyclic broadband emissions that can be picked up with the aid of high-sensitivity AE sensor placed on the outer pipe surface. Such sensors are being increasingly applied for the characterization of multiphase flow systems [1].

Some research has demonstrated AE signal characteristics such as dominant frequency components are a function of particle size and therefore support the use of spectral analysis for granulometry estimation applications [2]. Most publications focus on achieving the mass flow rate classification rather than actual continuous prediction, thereby limiting their potential for practical implementation [3]. Also, one of suggested methods was to estimate mass flow rate based on counting the peaks of the time-domain signal [4]; however, this is highly susceptible to the threshold value which is selected manually, compromising its validity. Additionally, regression methods that assess the root mean square (RMS) values of predefined frequency ranges have also been suggested [5,6], providing a more quantitative solution. The problem of classification has been tackled through the application of empirical mode decomposition (EMD) [7,8] and convolutional neural networks (CNNs) [8,9], both striving to enhance the feature extraction from nonstationary acoustic emission signals. There are other researchers who have attempted to derive analytical polynomial models that utilize empirically determined coefficients, thus providing a simpler yet interpretable method of connecting AE features with physical flow properties [9,10]. Despite the promising results, existing AE-based approaches are limited in their use. Most are restricted to empirical correlations or rely on simple signal properties (like RMS and kurtosis), which could be non-generalizable to various regimes of flow. In addition, there is presently no comprehensive model that simultaneously predicts both the solid mass flow rate and the particle size class from the AE data – two related parameters that are required for optimal process characterization.

The objective of the current study is to create an economic, non-invasive technique for the simultaneous determination of solid mass flow rates and solid particle size classification from acoustic emission signals via a clamp-on MEMS microphone. We present a signal processing framework comprising time-frequency analysis and some statistical features estimation, followed by their application to regression and classification modeling to analyze acoustic emission signatures at different flow conditions.

Section 2 outlines the methodology, covering data processing pipelines used on both the classification and regression tasks, as well as evaluation protocols for the predictive models. Section 3 describes the experimental apparatus, such as the design of the testbed, the architecture of the data acquisition system, the classification of the classes of sand, and the method employed in calculating the theoretical solid mass flow

rate. Section 4 is attributed to results and discussion, where the results of the classification and regression models are presented alongside a comprehensive analysis of their performance.

Methodology

The core idea of this work is to capture and analyze acoustic emission (AE) signals generated by a water-sand flow in a closed-loop laboratory setup. The installation imitates real conditions where solid particles are transported through a pipeline by the liquid phase.

At each experimental run, a fixed amount of sand was introduced into the circulating system, with fluid velocity kept constant. The flow entered the measurement zone where collisions between particles and pipe walls caused strong acoustic activity. AE signals were captured by a clamp-on MEMS microphone working at 100 kHz sampling rate. Each signal was recorded for about 10 seconds and saved as plain ASCII text. Later, these files were decoded and organized into a structured database containing the raw acoustic time series. The recordings were then processed through two digital analysis pipelines: Wavelet decomposition (DWT) and Spectral analysis (PSD).

Spectral analysis was used to estimate the power spectral density (PSD) of the AE signals, which later served as input to regression models predicting the solid mass flow rate. For the classification task – identifying sand type – features were extracted using discrete wavelet transform (DWT).

Classification Method

The classification approach used in this work is adapted from a previous study [11] focused on analyzing acoustic emission during wood fracture under three-point bending. They applied a combination of DWT to the AE signals, extracting statistical features to classify the fracture phases. Their method achieved an F1-score of 0.945 and accuracy exceeding 90%. Inspired by these results, we apply a similar strategy to classify sand types in a multiphase flow environment based on AE signals. The following sections detail the algorithm used in our implementation.

Overview of the Classification Algorithm. From each raw AE recording, five fixed-length subsamples are extracted. Each subsample is then processed using DWT. From the resulting components, a range of statistical descriptors is calculated – entropy, crest factor, impulse factor, margin factor, and waveform factor. These features serve as input to a linear classifier that assigns each subsample to a corresponding sand class.

The classification task is defined as a five-class problem. Class 0 represents the reference condition of clean water flow, free of any solid particles. Classes 1 through 4 correspond to four distinct sand types with different grain size distributions. Including the "no sand" class strengthens the model's capability to detect and classify both the presence and type of particulate matter, which is crucial for real-time flow monitoring and early warning of process deviations.

Wavelet Decomposition (DWT). Wavelet decomposition offers a multiresolution representation of signals in both time and frequency domains. Unlike the Fourier Transform, which employs globally supported sine and cosine waves, wavelets are time-localized and therefore more appropriate for the analysis of transient and non-stationary signals. This work utilizes the Discrete Wavelet Transform (DWT) using the Daubechies 4 ("db4") wavelet. The signal is processed through an iterative series of a pair of low-pass and high-pass filters and downsampling. An 8-level decomposition is used, yielding a detailed hierarchy of the signal's frequency components. This representation allows to isolate signal characteristics that are relevant for sand classification. At each level:

- Approximation coefficients (cA_N) represent the low-frequency content.
- Detail coefficients (cD_i) capture high-frequency variations at scale i .

Statistical Features Extraction. After decomposition, each signal wavelet component is characterized by a set of statistical descriptors:

- Entropy
- Crest Factor (ratio of maximum signal ripple to RMS of array)
- Impulse Factor (ratio of maximum signal ripple to average absolute value)
- Margin Factor (ratio of maximum signal ripple to root amplitude)
- Waveform Factor (ratio of RMS of array to rectification mean value)

These features are calculated for all relevant components of DWT and serve as the feature space for the classifier. The model is trained to map these features to one of the predefined sand classes, each corresponding to a specific characteristic grain size.

Sand Mass Flow Rate Estimation via Regression Method

In addition to classifying the sand type, this study addresses the task of estimating the mass flow rate of sand in a multiphase flow system using regression. The proposed approach is based on spectral analysis of acoustic emission (AE) signals.

Spectral Analysis (PSD). Spectral analysis is used to describe the frequency characteristics of a stationary signal. The power spectral density (PSD), estimated using Welch's method, reflects how the signal's energy is distributed across frequencies. PSD was selected as a feature source for regression due to its ability to compactly represent signal energy and highlight flow-related acoustic patterns.

RMS of Band-wise PSD and Factors Affecting Data Reliability. To estimate sand mass flow rate, we calculate the root-mean-square (RMS) of the PSD magnitude across fixed frequency bands. Since acoustic energy correlates with particle concentration, higher flow rates lead to more particle-wall impacts, generating stronger AE and higher PSD values. However, this increase is uneven across frequencies. Analyzing energy in specific bands is more effective, as different bands respond differently to flow changes. Using the full PSD, often with thousands of points, is impractical for regression due to high dimensionality. To reduce the input of prediction model, the spectrum is split into 11 bands, and RMS is computed over each: [0–0.5; 0.5–5; 5–10; 10–15; 15–20; 20–25; 25–30; 30–35; 35–40; 40–45; 45–50] kHz. An example of PSD profiles and band segmentation is shown in Figure 1.

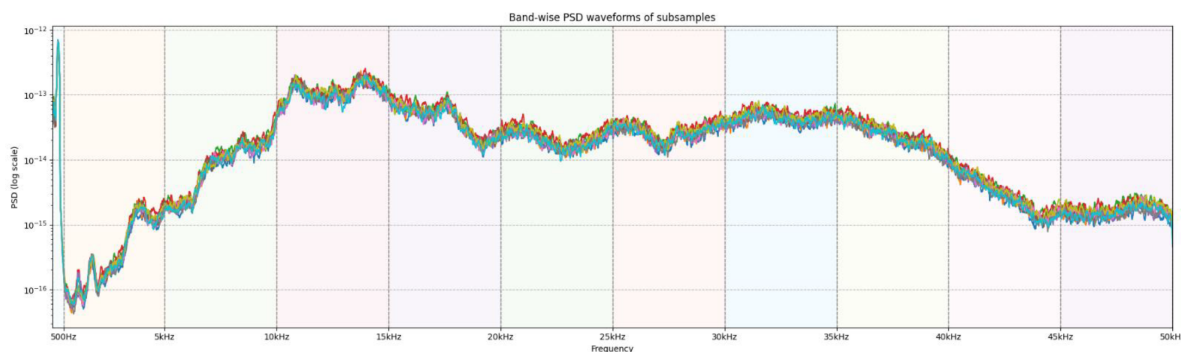


Figure 1—PSD curves for one of the signal's subsamples set.

Model for Regression and Classification

Gradient Boosting with default settings is applied to both classification and regression tasks. The algorithm builds decision trees sequentially, each correcting the errors of the previous one – fitting gradients of the

loss function in classification and residuals in regression. For classification, the model minimizes log-loss to output class probabilities; for regression, it minimizes the Mean Squared Error (MSE).

Classification Metrics. The classification performance is evaluated using the F1 score, the harmonic mean of precision and recall. Precision shows how many predicted samples of a class are correct, while recall measures how many actual class instances were correctly identified. The F1 score is calculated as (1).

$$F1 = \frac{2TP}{2TP + FP + FN} \quad (1)$$

where TP is true positives, FP is false positives and FN is false negatives.

F1 scores are calculated for every class in the multiclass setting, and the micro-averaging technique is used to average them. Also, the confusion matrix is used. It enhances the F1 score by assisting the identification of systematic misclassifications and inter-class confusions.

Regression Metrics. Regression performance is evaluated using residuals and the Mean Average Error (MAE). The residual for each data point is defined as (2).

$$r_i = y_i - \hat{y}_i \quad (2)$$

where y_i is the true value and $\hat{y}_i$ is model's prediction. Residuals indicate the absolute prediction error, which is especially relevant for low flow rates where even small errors may be critical.

The MAE is computed as the average of absolute values of all residuals:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (3)$$

MAE was selected to evaluate model performance because, unlike relative error metrics, it remains well-defined and meaningful even when the true values include zeros, avoiding instability or division errors.

Despite this, the average relative error is also computed for evaluation of model performance. Should be noted, that this metric does input only the non-zero entries. The expression for ARE evaluation is following:

$$ARE = \frac{1}{N} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \quad (4)$$

Experimental setup

Description of experimental setup

The experiment uses a closed hydraulic loop to circulate water with solids. Figure 2 shows the system scheme. At the start, tap water is manually added to the main reservoir, which has a conical bottom to help suspend solids and condition flow. A submersible pump with power output regulation drives the flow. Normally, the flow direction valve sends fluid back to the reservoir, bypassing the hydrocyclone, closing the loop. The defined quantity of solids is added directly into the reservoir. The conical bottom and inlet hoses create a strong vortex, keeping solids evenly mixed and pushing them toward the pump intake at the center bottom. This prevents sedimentation and keeps solid concentration stable. The pump draws the mixture through a long straight pipe where flow stabilizes near laminar. Then the flow passes a bent measurement segment designed to increase particle-wall interaction before returning to the reservoir. After the experiment, the valve switches flow to the hydrocyclone, which is intended to separate solids from water. Pure water is discarded; sand goes to a solids reservoir for drying and reuse.

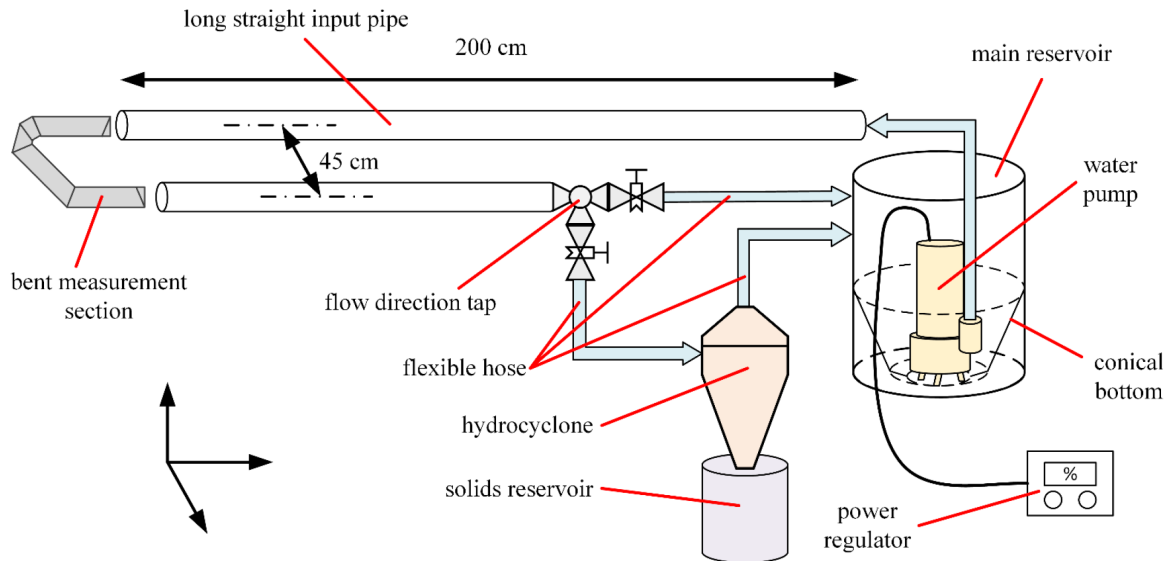


Figure 2—Experimental setup diagram.

Description of data acquisition system

The data acquisition system architecture is in Figure 3. It includes a MEMS microphone connected to an ESP32 microcontroller via PDM interface. The microphone is fixed to the pipe wall by a custom plastic clip for stable position and acoustic contact. Air fills the gap between pipe and microphone sensor. Acoustic signals go from ESP32 to host PC wirelessly using UDP over Wi-Fi router. Sampling rate is set before experiment by desktop app. After connection, ESP32 streams microphone data to PC in real time. Experiment control (recording start and stop) is made manually via the app interface.

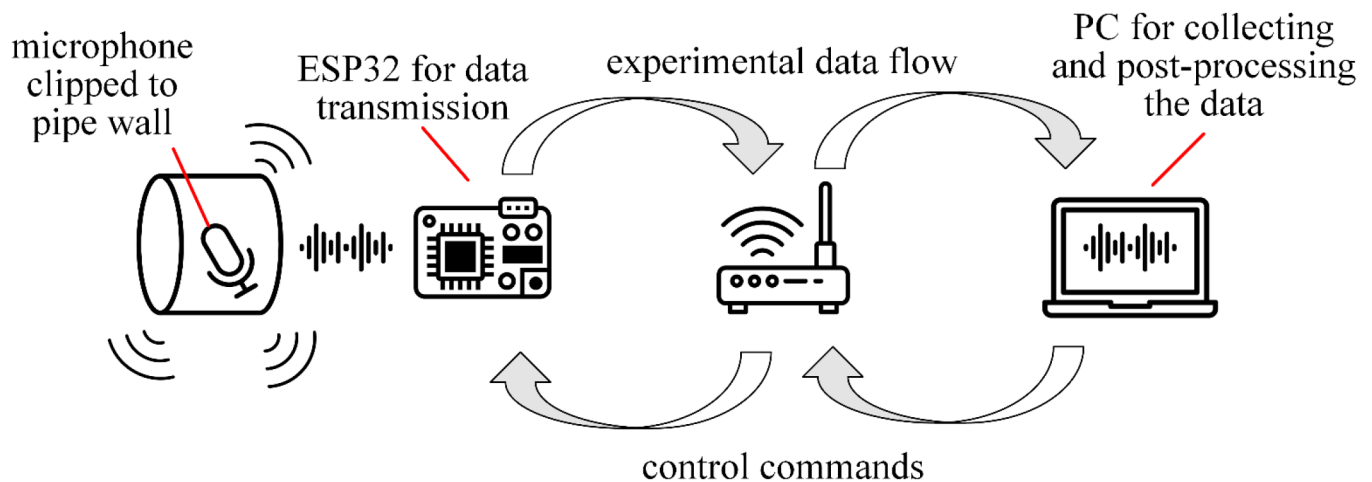
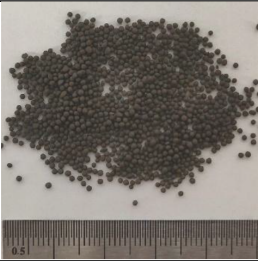


Figure 3—Data acquisition system.

Description of sand types

In presented study, there are 4 types of sand, which are utilized to test the system capabilities to predict the characteristic grain size. Table 1 accumulates the main parameters of sands.

Table 1—Description of sand classes.

				
class	1	2	3	4
name	rounded sand	well-rounded aluminum silicate spheres	angular sand	angular sand
size	1 – 2 mm	0.5 – 1 mm	4 – 8 mm	2 – 4 mm

Linking mass of diluted solids to mass flow rate

In this study, variable solid mass flow rates are controlled by changing the amount of solids added and adjusting pump output. The procedure to find solid mass flow rate is:

1. Create empirical calibration curve relating RMS voltage of loaded pump to power regulator percentage.
2. Assume pump power is proportional to squared RMS voltage.
3. Use calibration to link regulator percentage with actual power; vary power in 10% increments during experiments.
4. Estimate flow rate by linear interpolation from nominal pump capacity at 100% power for each step.
5. Knowing system volume and assuming uniform solid concentration, calculate solid mass flow rate by formula:

$$\dot{m}_s = \dot{m}_w \cdot \frac{c_s}{\rho_w}, \quad (5)$$

where $\dot{m}_s$ is target solid mass flow rate in multiphase flow, $\dot{m}_w$ is mass flow rate (productivity) of the pure system without solids, c_s is solids concentration in system, equals to ratio of diluted solids mass over volume of system, ρ_w is mass density of water, 1000 kg/m³

Results

Collection of AE samples

The complete set of experimental data consists of 397 recordings each being split to 5 subsamples for increasing the dataset volume. The total amount of data samples is 1985. AE recordings were made for variable sand classes and sweep of solid mass flow rate. The sweep of mass flow rate during the experiment can be represented as in [Figure 4](#) with minimum value of 0 and maximum of 250 g/liter.

Flow rate of solids, g/s		velocity of flow, cm/s										
		0	31.572	63.144	94.716	126.29	157.86	189.43	221	252.58	284.15	315.72
		pump power, %										
		<29%	30%	36%	40%	44%	48%	53%	56%	59%	64%	79%
solids concentration, g/liter	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	2.50	0.00	0.50	1.00	1.50	2.00	2.50	3.00	3.50	4.00	4.50	5.00
	5.00	0.00	1.00	2.00	3.00	4.00	5.00	6.00	7.00	8.00	9.00	10.00
	6.25	0.00	1.25	2.50	3.75	5.00	6.25	7.50	8.75	10.00	11.25	12.50
	10.00	0.00	2.00	4.00	6.00	8.00	10.00	12.00	14.00	16.00	18.00	20.00
	12.50	0.00	2.50	5.00	7.50	10.00	12.50	15.00	17.50	20.00	22.50	25.00
	18.75	0.00	3.75	7.50	11.25	15.00	18.75	22.50	26.25	30.00	33.75	37.50
	20.00	0.00	4.00	8.00	12.00	16.00	20.00	24.00	28.00	32.00	36.00	40.00
	25.00	0.00	5.00	10.00	15.00	20.00	25.00	30.00	35.00	40.00	45.00	50.00
	37.50	0.00	7.50	15.00	22.50	30.00	37.50	45.00	52.50	60.00	67.50	75.00
	50.00	0.00	10.00	20.00	30.00	40.00	50.00	60.00	70.00	80.00	90.00	100.00
	62.50	0.00	12.50	25.00	37.50	50.00	62.50	75.00	87.50	100.00	112.50	125.00
	75.00	0.00	15.00	30.00	45.00	60.00	75.00	90.00	105.00	120.00	135.00	150.00
	100.00	0.00	20.00	40.00	60.00	80.00	100.00	120.00	140.00	160.00	180.00	200.00
	125.00	0.00	25.00	50.00	75.00	100.00	125.00	150.00	175.00	200.00	225.00	250.00
liquid flow rate, liter/s		0	0.2	0.4	0.6	0.8	1	1.2	1.4	1.6	1.8	2

Figure 4—Mass flow rate sweep.

Results of classification

In result of solving the classification problem, the following insights can be drawn:

1. The classification model demonstrates high accuracy in distinguishing between different sand types and the absence of sand. The confusion matrix (Figure 5) shows particularly strong performance for classes 1 through 4, with each achieving over 90% correct predictions.
2. Misclassifications are generally limited to adjacent classes, suggesting the model captures the gradation in sand grain sizes. Notably, the model correctly identifies "no sand" (class 0) in 91 out of 96 cases, reflecting strong specificity and supporting its utility in detecting both presence and type of particulate matter in multiphase flow.
3. The F1-score of multilabel classification reaches value of 0.917 when test data are half of the total dataset.

		Confusion Matrix				
True	0	73	0	0	7	0
	1	1	199	13	0	12
	2	0	11	213	0	1
	3	13	0	0	203	9
	4	5	3	0	7	223
		0	1	2	3	4
		Predicted				

Figure 5—Confusion matrix of solid type classification.

Results of regression

In result of solving the regression problem, the following insights can be drawn:

1. Residual analysis (Figure 6) shows prediction errors symmetrically spread around zero over full target range, meaning low bias. Logarithmic colormap reveals most residual-to-actual ratios stay below 10% at mid and high flow rates, showing good relative accuracy. Larger relative errors appear at low true values, as expected, but absolute residuals stay small. This confirms the model generalizes well across wide range with good absolute precision.
2. The model exhibits elevated average relative error (ARE) and mean absolute error (MAE) in low-rates region (Figure 7). This trend may be attributed to the high sensitivity of relative error at small magnitudes and potential feature noise or resolution limitations in the low-value range. Small absolute deviations can produce disproportionately high relative errors, particularly when true values approach zero. In contrast, predictions for higher true values – where data is sparse – exhibit more stable and lower error levels, likely due to better separation between classes and reduced penalization in relative terms.
3. The observed behavior suggests that the model does not yet fully exploit the information present in low-value samples. This can be addressed by enhancing feature sensitivity in the low range, applying noise reduction techniques, or revising the loss function (e.g., combining MAE with relative error metrics or adopting a Huber loss). Additionally, training with value-dependent weighting or transforming the target variable may help reduce bias toward high-value regions.

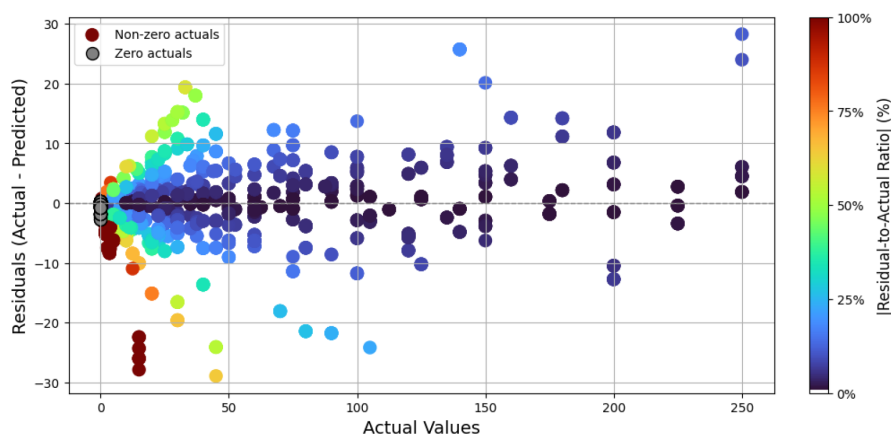


Figure 6—Residuals of individual predictions vs. Actual Values.

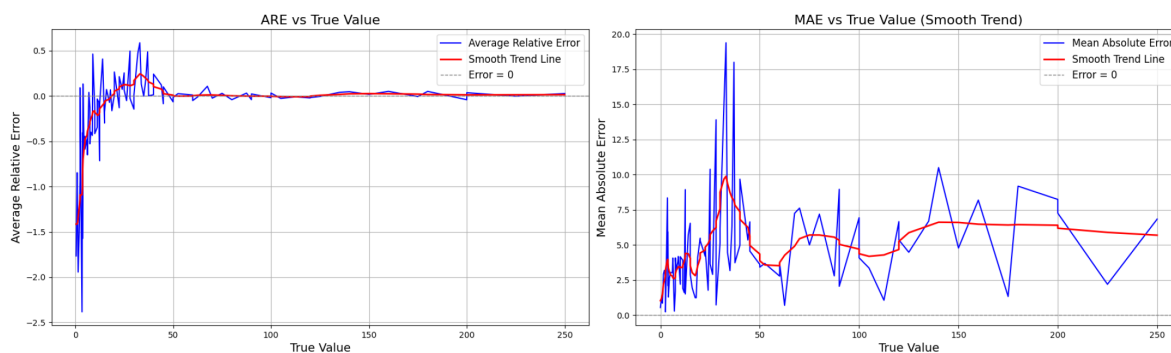


Figure 7—Analysis of ARE (left) and MAE (right) vs. True values.

Conclusion

The proposed model performs well in regression and classification of acoustic emissions from multiphase flow. Regression shows predicted solid mass flow rates close to actual values, with low residuals at medium and high flow rates. Relative errors rise at low flows due to noise and skewed data but without strong bias, indicating model robustness.

Classification accurately identifies sand presence and type. The confusion matrix shows reliable distinction between no sand and four sand classes. Most errors happen between neighboring classes, likely due to gradual grain size changes rather than clear signal differences. High true positive rate for no sand confirms good detection of clean flow, important for real-time alerts.

These results demonstrate acoustic emission analysis as a low-cost, non-invasive monitoring method for multiphase systems. Future work should balance dataset classes better and test under different pipe geometries and flow rates.

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